

Saratchandra Patnaik

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Immediately available to work

Cloud-native software engineer with production experience in Kubernetes (AWS EKS), Docker, and GitOps (ArgoCD). Skilled in managing end-to-end deployment lifecycles and engineering scalable backend solutions using Python and FastAPI. Innovative in building GenAI-driven observability pipelines for automated log analysis, with a strong foundation in hybrid/edge systems, Linux, and reliability-focused troubleshooting.

EDUCATION

Master of Science in Computer Science (CGPA: 3.80/4)

Graduated December 2025

Arizona State University, Tempe, AZ

Bachelor of Technology in Computer Science (CGPA: 9.11/10)

Graduated May 2022

Christ University School of Engineering, Bengaluru, India

PROFESSIONAL EXPERIENCE

Software Engineer: Amagi Media Labs, India [AWS EKS, ArgoCD, FAST API]

August 2022 - November 2023

- Deployed end-to-end software development lifecycles for cloud-native microservices on AWS EKS, coordinating across cross-functional teams to ensure seamless integration and 99.9% release stability.
- Operated and scaled a Kubernetes-based microservices platform hosting 15+ services, increasing deployment frequency by 30% through robust containerization and reliability practices.
- Implemented GitOps CI/CD workflows using ArgoCD to standardize rollout procedures, significantly reducing deployment risks and accelerating recovery times during critical regressions.
- Engineered high-performance RESTful APIs using FastAPI and Python to support backend operations, optimizing asynchronous request handling and enhancing data flow efficiency across internal tools.
- Designed an AI-driven observability pipeline that ingests server logs into Large Language Models, automating root cause analysis and slashing manual debugging time for complex incidents by 50%.
- Automated customer onboarding and integrations using Terraform and Python workflows, cutting manual setup efforts by 90% and reducing CDN delivery failures by 95%.

Software Intern: Blueplanet Solutions Inc., India [MySQL]

April 2021 - June 2021

- Optimized MySQL database using stored procedures for 'Campus Club', improving query speed by 50% and enabling faster data access for 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, JavaScript, and PHP, reducing search time by 60% and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Real-Time Face Recognition on AWS Greengrass [MQTT, AWS Lambda]

- Engineered a resilient edge-to-cloud pipeline using AWS IoT Greengrass, MQTT, and Lambda/SQS, automating deployment workflows via shell scripts to successfully pass 100+ real-time autograder test cases.
- Deployed MTCNN and FaceNet models under strict pip-free constraints by integrating raw PyTorch source, optimizing runtime efficiency within AWS Greengrass limits and debugging distributed components via CLI/logs.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, REST APIs

Databases: MySQL, PostgreSQL